

■ MATHEMATICS · GRADE 6

# Linear Programming

## Graphical Method

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Plot · Shade · Optimise

- Duration: 30 Minutes
- Level: Beginner
- Planrs Academy

Lesson at a Glance

| Subject | Topic                                 | Grade | Duration   | Level    |
|---------|---------------------------------------|-------|------------|----------|
| Maths   | Linear Programming — Graphical Method | 6     | 30 Minutes | Beginner |

Learning Objectives

| #   | By the end of this lesson, students will be able to...                                         |
|-----|------------------------------------------------------------------------------------------------|
| LO1 | Define linear programming and explain decision variables, constraints, and objective function. |
| LO2 | Write real-life constraints as linear inequalities.                                            |
| LO3 | Plot constraint lines on a coordinate graph and shade the correct half-plane.                  |
| LO4 | Identify the feasible region by overlapping all constraints.                                   |
| LO5 | Apply the corner-point method to find the optimal solution.                                    |
| LO6 | Solve a school timetable optimisation problem from scratch.                                    |

1. What is Linear Programming?

Linear programming is a mathematical method for finding the **best possible outcome** (maximum profit, minimum cost, most items) given a set of rules called **constraints**. It uses a coordinate graph to turn real-world decisions into a visual puzzle.

★ ■ Example: Your school canteen makes Idlis (x) and Samosas (y) every morning.  
 The chef has limited ingredients and oven time — these are the **CONSTRAINTS**.  
 The goal is to maximise total items sold — this is the **OBJECTIVE FUNCTION**.

| Part                         | What It Means                                   | Indian Example                           |
|------------------------------|-------------------------------------------------|------------------------------------------|
| Decision Variables (x and y) | The quantities we want to decide on.            | x = Idlis, y = Samosas made per day      |
| Constraints                  | Real-world limits written as inequalities.      | $x + y \leq 10$ (max 10 items per batch) |
| Objective Function           | The expression we want to maximise or minimise. | Maximise $Z = x + y$ (total items sold)  |

## ■ 2. Understanding Constraints

Constraints are **rules or limits** written as mathematical inequalities. They tell us what is allowed and what is not. Every constraint has two sides — the **boundary line** and the **feasible half-plane**.

| Symbol               | Meaning               | Line Type         | Example         |
|----------------------|-----------------------|-------------------|-----------------|
| $\leq$ or $\geq$     | Includes the boundary | Solid line ■■■    | $x + y \leq 10$ |
| $<$ or $>$           | Excludes the boundary | Dashed line ■ ■ ■ | $x + y < 10$    |
| $x \geq 0, y \geq 0$ | No negative values    | Axes themselves   | Always applies  |

## ■ 3. Graphing Constraints — Step by Step

To graph a linear constraint, follow these four steps every time. We will use  $x + y \leq 10$  as our example.

1

### Convert to Equation

Replace the inequality sign with = to get the boundary line.  $\rightarrow x + y = 10$

2

### Find Two Points

Set  $x = 0 \rightarrow y = 10$ , giving point (0, 10). Set  $y = 0 \rightarrow x = 10$ , giving point (10, 0).

3

### Draw the Line

Plot (0, 10) and (10, 0) on the graph. Connect them with a solid line (since we have  $\leq$ ).

4

### Shade the Correct Side

Test point (0, 0):  $0 + 0 = 0 \leq 10 \rightarrow \text{TRUE}$ . Shade the side containing (0, 0) — below the line.

## ■ 4. The Feasible Region

When we plot ALL constraints on the same graph, the **overlapping shaded area** that satisfies every constraint simultaneously is called the **Feasible Region**. The optimal solution always lies at one of its **corner points (vertices)**.

### ■ Canteen Problem

Variables:  $x$  = Idlis,  $y$  = Samosas

#### Constraints:

①  $x + y \leq 10$  (total batch limit)

②  $2x + y \leq 14$  (oven time limit)

③  $x \geq 0, y \geq 0$

#### Corner Points:

$(0, 0) \cdot (7, 0) \cdot (4, 6) \cdot (0, 10)$

■ The optimal answer is at one of these four yellow dots!

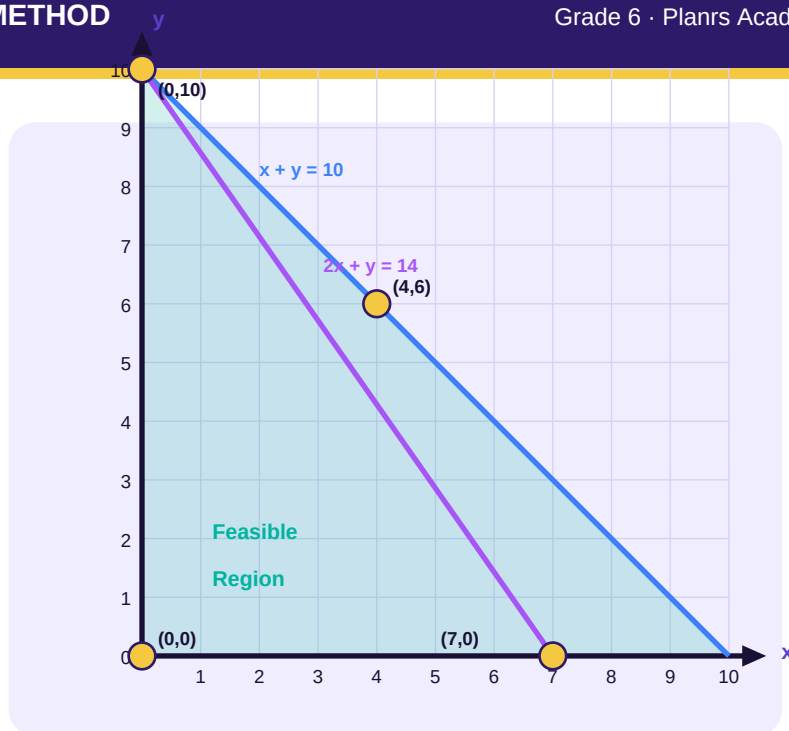


Figure 1 — Feasible region (teal shading) for the canteen problem. Corner points shown in gold.

## ■ 5. The Corner Point Method

**Fundamental Theorem of Linear Programming:** The optimal value of the objective function (maximum or minimum) always occurs at one of the **corner points (vertices)** of the feasible region.

**Objective Function: Maximise  $Z = x + y$  (total items sold)**

| Corner Point | x value | y value | $Z = x + y$ | Result     |
|--------------|---------|---------|-------------|------------|
| $(0, 0)$     | 0       | 0       | 0           | —          |
| $(7, 0)$     | 7       | 0       | 7           | —          |
| $(4, 6)$     | 4       | 6       | 10          | — check! — |
| $(0, 10)$    | 0       | 10      | 10          | ■ MAXIMUM  |

✓ ■ Maximum  $Z = 10$  at corners  $(4, 6)$  and  $(0, 10)$ .

The canteen should make 0 idlis + 10 samosas OR 4 idlis + 6 samosas to maximise output!

## ■ 6. School Timetable Optimisation — Full Problem

Let's apply everything we have learnt to a complete Indian school problem!

- A school has  $x$  = Maths periods and  $y$  = Science periods per week.  
Objective: Maximise  $Z = x + 2y$  (Science counts double for exam prep)

### Constraints:

| Label | Inequality           | Plain English                             |
|-------|----------------------|-------------------------------------------|
| C1    | $x + y \leq 10$      | Maximum 10 periods total per week         |
| C2    | $x \geq 3$           | At least 3 Maths periods (CBSE minimum)   |
| C3    | $y \geq 2$           | At least 2 Science periods (CBSE minimum) |
| C4    | $x \geq 0, y \geq 0$ | No negative periods (common sense!)       |

### Step 1 — Find the corner points of the feasible region:

Solving the constraint system gives us three corner points: **(3, 2)**, **(8, 2)**, and **(3, 7)**.

### Step 2 — Evaluate the objective function at each corner:

| Corner Point | $x$ | $y$ | $Z = x + 2y$ | Calculation           |
|--------------|-----|-----|--------------|-----------------------|
| (3, 2)       | 3   | 2   | 7            | $3 + 2 \times 2 = 7$  |
| (8, 2)       | 8   | 2   | 12           | $8 + 2 \times 2 = 12$ |
| (3, 7)       | 3   | 7   | 17           | $3 + 2 \times 7 = 17$ |

- ★ ■ **OPTIMAL SOLUTION:** Maximum  $Z = 17$  at corner point (3, 7).  
→ Schedule 3 Maths periods + 7 Science periods for the best exam preparation!  
This satisfies ALL constraints and gives the highest objective value.

## 7. 6-Step Solution Checklist

| Step          | Task                         | How To Do It                                   |
|---------------|------------------------------|------------------------------------------------|
| <b>Step 1</b> | Identify Decision Variables  | Name $x$ and $y$ — what are you deciding?      |
| <b>Step 2</b> | Write the Constraints        | Convert each rule into an inequality           |
| <b>Step 3</b> | Plot Constraint Lines        | Convert to $=$ → find 2 points → draw the line |
| <b>Step 4</b> | Shade & Find Feasible Region | Test (0,0) to shade; overlap all constraints   |

|        |                    |                                              |
|--------|--------------------|----------------------------------------------|
| Step 5 | List Corner Points | Mark all vertices of the feasible region     |
| Step 6 | Evaluate & Choose  | Substitute each corner into Z; pick the best |

⇒ 8. Practice Problems

Problem A — Canteen Budget ■ (Easy)

⇒ You have ₹20 to spend on pencils (x, ₹3 each) and erasers (y, ₹2 each).

Objective: Maximise  $Z = x + y$  (total items purchased)

Constraint 1:  $3x + 2y \leq 20$  (budget limit)

Constraint 2:  $x \geq 0, y \geq 0$  (no negatives)

TASK: Plot the constraint, shade the feasible region, and find the optimal solution.

Problem B — Diwali Mela Stall ■■ (Medium)

⇒ A student has 2 hours (120 min) and ₹200 for a Diwali mela stall.

Diyas (x): 10 min each, costs ₹5 each.

Rangoli Kits (y): 20 min each, costs ₹20 each.

Objective: Maximise  $Z = x + y$  (total items made)

Constraints:  $10x + 20y \leq 120$  (time) ·  $5x + 20y \leq 200$  (budget) ·  $x, y \geq 0$

TASK: Graph both constraints, find the feasible region, and use corner points to solve.

Problem C — Cricket Team ■■■ (Challenge)

■ A school cricket team selects batsmen (x) and bowlers (y).

Total players must be exactly 11:  $x + y = 11$

Need at least 5 batsmen:  $x \geq 5$     Need at least 4 bowlers:  $y \geq 4$

Objective: Maximise batting score =  $10x + 7y$

TASK: Find ALL corner points of the feasible region and determine the optimal team!

■ 9. Key Terms Glossary

| Term               | Definition                                                                 |
|--------------------|----------------------------------------------------------------------------|
| Linear Programming | A method to find the best outcome (max/min) subject to linear constraints. |

|                                  |                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------|
| <b>Decision Variables</b>        | The unknowns (x, y) whose values we are trying to determine.                          |
| <b>Constraint</b>                | A rule expressed as a linear inequality limiting the variables.                       |
| <b>Objective Function</b>        | The mathematical expression we want to maximise or minimise (e.g., $Z = x + 2y$ ).    |
| <b>Feasible Region</b>           | The overlapping area on the graph that satisfies ALL constraints simultaneously.      |
| <b>Corner Point (Vertex)</b>     | A point where two constraint boundaries intersect at the edge of the feasible region. |
| <b>Optimal Solution</b>          | The corner point that gives the best value of the objective function.                 |
| <b>Non-negativity Constraint</b> | The requirement that $x \geq 0$ and $y \geq 0$ (variables cannot be negative).        |

## ■ 10. Real-World Connections (Indian Context)

Linear programming is used every single day in India across many industries. Even the government uses it for planning!

| Icon | Application                    | How LP Helps                                                             |
|------|--------------------------------|--------------------------------------------------------------------------|
| ■    | <b>CBSE School Timetabling</b> | Scheduling subjects within weekly period limits and teacher availability |
| ■    | <b>Mid-Day Meal Planning</b>   | Meeting nutrition standards at minimum cost for school meals             |
| ■    | <b>Cricket Team Selection</b>  | Optimising runs scored within squad, batting/bowling balance constraints |
| ■    | <b>Shop Stock Planning</b>     | Maximising profit from limited shelf space and supplier quantities       |
| ■    | <b>School Bus Routing</b>      | Covering all routes with minimum number of buses and drivers             |

## ■ 11. Common Mistakes to Avoid

| Mistake                  | How to Fix It                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------|
| ■ Shading the wrong side | ■ Always test the point (0, 0) first. If it satisfies the inequality → shade that side. |

|                             |                                                                                                                          |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| ■ Missing non-negativity    | ■ ALWAYS include $x \geq 0$ and $y \geq 0$ . They are invisible constraints that limit the region to the first quadrant. |
| ■ Forgetting a corner point | ■ Every intersection of two constraint lines (including axes) inside the region is a corner. Check them all!             |
| ■ Using only one constraint | ■ The feasible region comes from ALL constraints overlapping — never just one.                                           |
| ■ Not labelling axes        | ■ Always write what x and y represent on your graph. e.g., "x = Maths Periods"                                           |

## 12. Exit Ticket — Quick Check ➡

Answer these 3 questions before you leave class today!

| Q  | Question                                                                                                                                                                                 |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1 | What are the three components of every linear programming problem?                                                                                                                       |
| Q2 | If the constraint is $x + 2y \leq 8$ , draw the boundary line and shade the feasible half-plane. Does (0, 0) satisfy this constraint? Show your working.                                 |
| Q3 | The feasible region of a problem has corners at (2, 1), (5, 1), and (2, 4). The objective function is $Z = 3x + 2y$ . Find the maximum value of Z and state which corner point gives it. |

■ ■ Answer Key is available in the Teacher's Guide.

Success Criterion: Students should score 2/3 or above to proceed confidently.